Problem Set 1: The Geometry of Events

Worked Solutions

Sets, Venn diagrams, De Morgan's laws, and inclusion–exclusion

80 points + 10 optional challenge points

Purpose. Sets are the grammar in which probability will be written. This assignment develops fluency in translating among words, symbols, Venn diagrams, and counts. Give reasons for your answers: a diagram may support an argument, but it does not replace one when a proof is requested.
Notation. All complements are taken relative to the stated universe Ω. The symbol $|A|$ denotes the number of elements in a finite set A. The word "or" is inclusive unless explicitly stated otherwise.

Instructor note

The core assignment is Problems 1–8. Problem 3 checks visual translation; Problems 4 and 5 move from pictures to inclusion–exclusion; Problems 6–8 provide a gentle transition from examples to proof. The challenge can be assigned even if divisibility has not been discussed formally.

I. Sets in symbols and pictures

1. Membership and operations. [10 points] Let

$$\Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}, \quad A = \{2, 4, 6, 8, 10, 12\}, \quad B = \{2, 3, 5, 7, 11\}.$$

List the elements of each set.

(a) $A \cap B$

(b) $A \cup B$

(c) A^c

(d) $A \setminus B$

(e) $(A \cup B)^c$

Solution

The sets A and B are, respectively, the even elements and the prime elements of Ω. Directly checking membership gives

$$\begin{aligned} A \cap B &= \{2\}, \\ A \cup B &= \{2, 3, 4, 5, 6, 7, 8, 10, 11, 12\}, \\ A^c &= \{1, 3, 5, 7, 9, 11\}, \\ A \setminus B &= \{4, 6, 8, 10, 12\}, \\ (A \cup B)^c &= \{1, 9\}. \end{aligned}$$

Notice that 2 belongs to both A and B, while 1 and 9 belong to neither.

2. Negating carefully: De Morgan's laws. [10 points] Let $A, B \subseteq \Omega$.

(a) Translate $(A \cup B)^c = A^c \cap B^c$ into an English sentence.

(b) Prove the identity in part (a) by showing that an arbitrary element $x \in \Omega$ belongs to the left side exactly when it belongs to the right.

(c) State the other De Morgan law and prove it in the same way.

Solution

(a) "Being in neither A nor B is the same as being outside A and outside B." Equivalently, "not (A or B)" means "not A and not B."

(b) For any $x \in \Omega$,

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin A \cup B \\ &\iff (x \notin A) \text{ and } (x \notin B) \\ &\iff (x \in A^c) \text{ and } (x \in B^c) \\ &\iff x \in A^c \cap B^c. \end{aligned}$$

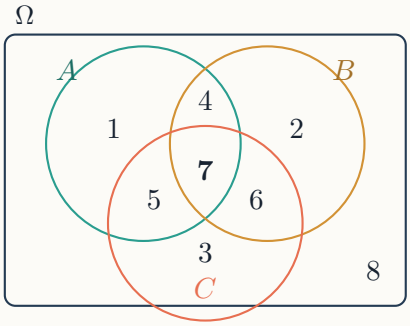
Since this holds for every x, the sets are equal.

(c) The other law is $(A \cap B)^c = A^c \cup B^c$. Indeed,

$$\begin{aligned} x \in (A \cap B)^c &\iff x \notin A \cap B \\ &\iff (x \notin A) \text{ or } (x \notin B) \\ &\iff (x \in A^c) \text{ or } (x \in B^c) \\ &\iff x \in A^c \cup B^c. \end{aligned}$$

3. Reading and shading Venn diagrams. **[12 points]**

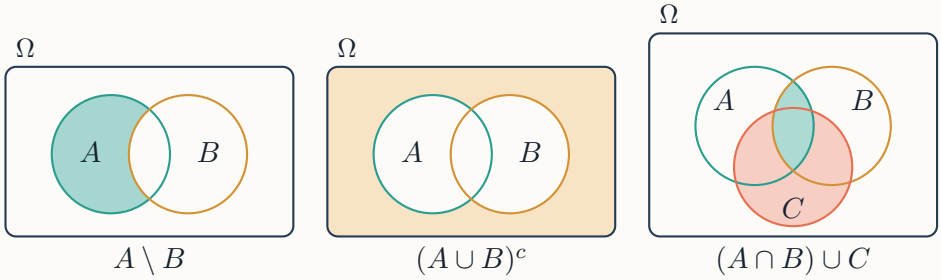
In the numbered diagram below, each number labels one of the eight disjoint regions determined by A, B, and C.



(a) Give the region number(s) belonging to each set:

$$A \cap B \cap C, \qquad A \setminus (B \cup C), \qquad (A \cup B)^c, \qquad (A \cap B) \setminus C.$$

(b) Shade the requested set in each diagram.



Solution

For part (a), the region lists are

set	regions
$A \cap B \cap C$	7
$A \setminus (B \cup C)$	1
$(A \cup B)^c$	3, 8
$(A \cap B) \setminus C$	4

The diagrams above show the required shading. In the third diagram, all of C is shaded, together with the lens $A \cap B$ (including any overlap that the lens already has with C).

II. Counting with overlaps

4. Two-set inclusion–exclusion. [10 points] Among 120 students, 68 study calculus, 54 study programming, and 31 study both. Assume every student is among the 120 being surveyed.

(a) How many study calculus or programming (or both)?

(b) How many study exactly one of the two subjects?

(c) How many study neither subject?

(d) Explain briefly why simply adding $68 + 54$ gives the wrong answer to part (a).

Solution

Let C be the calculus set and P the programming set.

(a) Inclusion–exclusion gives

$$|C \cup P| = |C| + |P| - |C \cap P| = 68 + 54 - 31 = 91.$$

(b) There are $68 - 31 = 37$ calculus-only students and $54 - 31 = 23$ programming-only students, hence $37 + 23 = 60$ who study exactly one subject.

(c) The number studying neither is $120 - 91 = 29$.

(d) In $68 + 54$, each of the 31 students in the overlap is counted twice. Subtracting the overlap once makes each such student count exactly once.

5. A three-set survey. [14 points] At a university with 200 surveyed students, let M, C, E denote the students taking mathematics, computing, and economics. Suppose

$$|M| = 105, \quad |C| = 90, \quad |E| = 75,$$
$$|M \cap C| = 50, \quad |M \cap E| = 40, \quad |C \cap E| = 35,$$
$$|M \cap C \cap E| = 20.$$

The pairwise intersection counts include students taking all three subjects.

(a) Determine the number in each of the eight regions of a three-set Venn diagram.

(b) How many students take at least one of the three subjects?

(c) How many take exactly one? Exactly two? All three? None?

(d) Check that your four answers in part (c) add to 200.

Solution

Start from the center. The triple intersection contains 20 students. Thus the pair-only regions have sizes

$$MC \text{ only} = 50 - 20 = 30, \quad ME \text{ only} = 40 - 20 = 20, \quad CE \text{ only} = 35 - 20 = 15.$$

Subtracting all overlap regions from each set gives

$$M \text{ only} = 105 - 30 - 20 - 20 = 35,$$

$$C \text{ only} = 90 - 30 - 15 - 20 = 25, \qquad E \text{ only} = 75 - 20 - 15 - 20 = 20.$$

Therefore the union has size

$$35 + 25 + 20 + 30 + 20 + 15 + 20 = 165,$$

so 35 students lie outside all three sets. Equivalently, three-set inclusion–exclusion gives

$$|M \cup C \cup E| = 105 + 90 + 75 - 50 - 40 - 35 + 20 = 165.$$

Exactly one subject: $35 + 25 + 20 = 80$. Exactly two: $30 + 20 + 15 = 65$. All three: 20. None: 35. Finally,

$$80 + 65 + 20 + 35 = 200,$$

as required.

6. Difference distributes over union. [10 points]

(a) Use De Morgan's law and the identity $A \setminus B = A \cap B^c$ to simplify $A \setminus (B \cup C)$.

(b) Prove directly, by following an arbitrary element x, that

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

(c) Explain the identity in one ordinary English sentence.

Solution

(a) We calculate

$$A \setminus (B \cup C) = A \cap (B \cup C)^c = A \cap (B^c \cap C^c) = (A \cap B^c) \cap (A \cap C^c),$$

where the final equality uses associativity and the fact that repeating A changes nothing. This is $(A \setminus B) \cap (A \setminus C)$.

(b) For any $x \in \Omega$,

$$\begin{aligned} x \in A \setminus (B \cup C) &\iff x \in A,\ x \notin B, \text{ and } x \notin C \\ &\iff x \in A \setminus B \text{ and } x \in A \setminus C \\ &\iff x \in (A \setminus B) \cap (A \setminus C). \end{aligned}$$

(c) The elements of A that are in neither B nor C are precisely those that remain after removing both B and C from A.

III. From diagrams to arguments

7. Cancellation can fail for sets. [6 points] The equation $a + c = b + c$ implies $a = b$ for numbers. Decide whether each proposed analogue for sets is true. Prove it if it is true; give a concrete

counterexample if it is false.

(a) If $A \cup C = B \cup C$, then $A = B$.

(b) If both $A \cup C = B \cup C$ and $A \cap C = B \cap C$, then $A = B$.

Solution

(a) This is false. Take $A = \{1\}$, $B = \{2\}$, and $C = \{1, 2\}$. Then $A \cup C = C = B \cup C$, but $A \neq B$.

(b) This is true. Let $x \in A$. If $x \in C$, then $x \in A \cap C = B \cap C$, so $x \in B$. If $x \notin C$, then $x \in A \cup C = B \cup C$; because $x \notin C$, we must have $x \in B$. Thus $A \subseteq B$. The same argument with A and B interchanged gives $B \subseteq A$, hence $A = B$.

8. Why three-set inclusion–exclusion works. [8 points] For finite sets A, B, C, prove

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Do this by choosing one element x and checking its total contribution to the right-hand side when x belongs to exactly zero, one, two, or three of the sets.

Solution

Consider a fixed element x. If x belongs to none of the sets, it contributes 0 to every term. If it belongs to exactly one set, it appears once among $|A|, |B|, |C|$ and nowhere else, so its contribution is 1. If it belongs to exactly two sets, it is added twice among the single-set terms and subtracted once in the corresponding pairwise intersection, contributing $2 - 1 = 1$. If it belongs to all three sets, it is added three times, subtracted three times, and added once in the triple intersection, contributing $3 - 3 + 1 = 1$. Thus every element of the union is counted exactly once and every element outside the union is counted zero times. The two sides are equal.

Optional challenge

9. Inclusion–exclusion among the integers. [10 optional points] How many integers from 1 through 500 are divisible by at least one of 4, 6, or 9? How many are divisible by none of them? Show how every term in your inclusion–exclusion calculation arises.

Solution

Let A_d be the set of integers from 1 through 500 divisible by d. Then $|A_d| = \lfloor 500/d \rfloor$. Intersections correspond to least common multiples:

set	A_4	A_6	A_9	$A_4 \cap A_6$	$A_4 \cap A_9$	$A_6 \cap A_9$	$A_4 \cap A_6 \cap A_9$
divisor	4	6	9	12	36	18	36
size	125	83	55	41	13	27	13

Therefore

$$|A_4 \cup A_6 \cup A_9| = 125 + 83 + 55 - 41 - 13 - 27 + 13 = 195.$$

The other $500 - 195 = 305$ integers are divisible by none of $4, 6, 9$.

Before submitting: check complements against Ω, include overlaps when you write "or," and make every inclusion–exclusion correction visible.